AIML MAJOR PROJECT

News Intelligence & Topic Discovery System

Unsupervised topic discovery in news articles using TF-IDF and K-Means

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THE PROBLEM

Manual sorting cannot keep up with changing news

Scale

Thousands of news articles appear every day.

Friction

Sorting them by hand is slow.

Drift

Topics change over time.

OBJECTIVES

- Clean a real news dataset.
- Turn text into numbers with TF-IDF.
- Discover topics with K-Means.
- Visualise clusters with PCA.
- Find similar articles with cosine similarity.

Key principle: Clustering is unsupervised — no category labels are used for training.

DATA PREPARATION

Clean text creates a usable learning input

RAW DATASET

2,225

BBC News full-text dataset

Business · entertainment · politics · sport · tech

DEDUPLICATION

98

duplicates removed

LEARNING INPUT

2,127

articles retained

A. Lowercase

B. Remove HTML and punctuation

C. Remove stop words and numbers

D. Keep names and abbreviations

No stemming keeps topic keywords readable. Dataset labels are kept only for evaluation.

SYSTEM WORKFLOW

One workflow connects cleaning, clustering, and discovery

A. News CSV B. Clean text C. TF-IDF vectors D. K-Means clusters E. Topic keywords and names



TF-IDF vectors → PCA 2D map



TF-IDF vectors → Cosine similarity search



K-Means clusters → Evaluation

End point: Streamlit dashboard with 5 pages.

MODEL COMPONENTS

Four methods turn text into explainable topics



TF-IDF

5,000 features; 1–2 word phrases; rare, article-relevant words gain weight.



K-Means

Groups similar articles into k topics; k = 5 by default; random seed 42.



PCA

Reduces thousands of dimensions to 2 for drawing; clustering does not use it.



Cosine similarity

Compares word usage and returns the top 5 similar articles.

Topic keywords: words that stand out most inside a cluster compared with all articles.

PRODUCT SURFACE

The dashboard turns the model into an explorable system

Five-page layout overview:

Overview

Topics, sizes, keywords, rename

Topic Visualization

Interactive PCA map; hover for titles

Model Evaluation

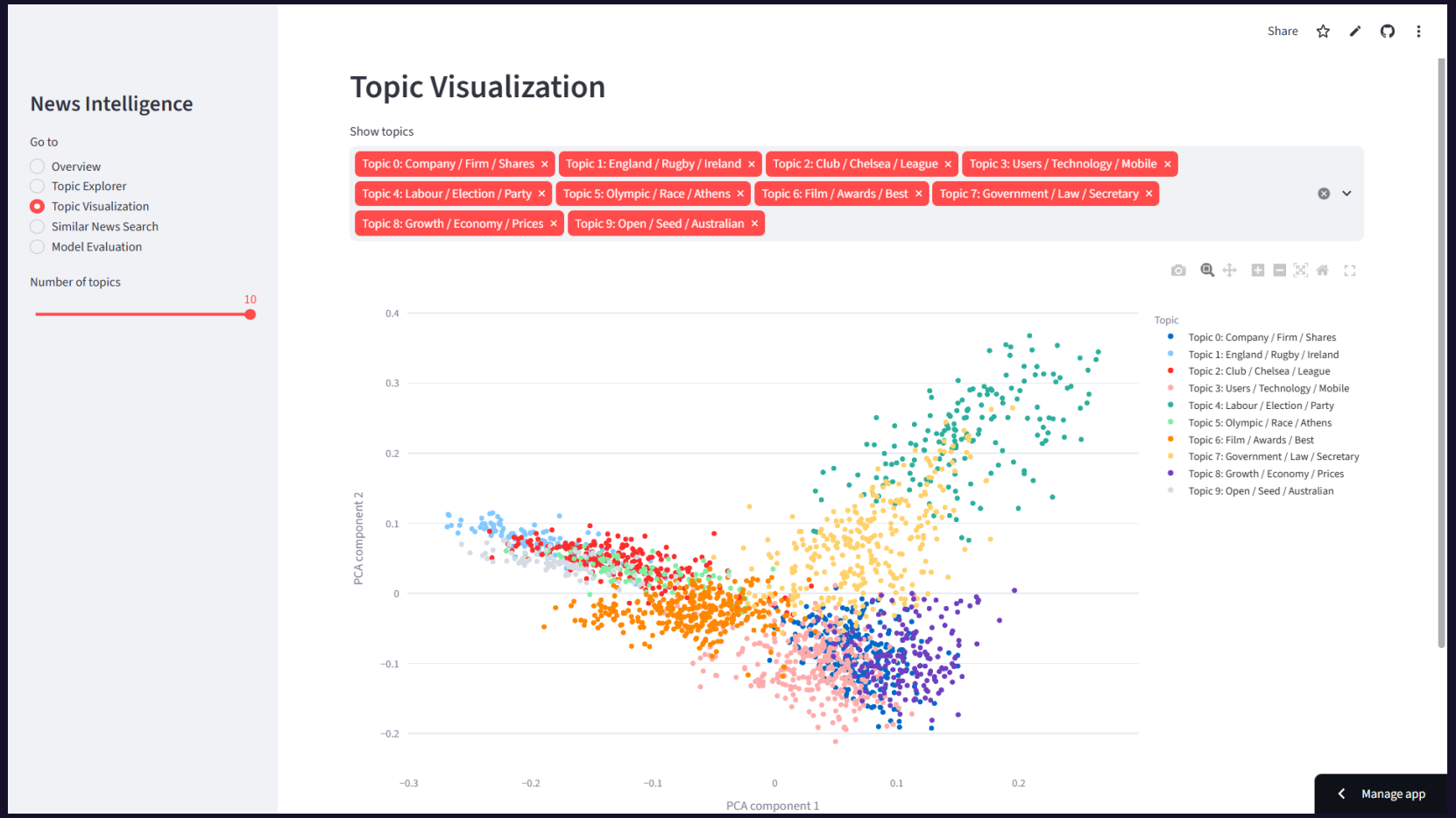
Inspect quality checks

Topic Explorer

Inspect discovered topics

Similar News Search

Choose an article or paste new text; top 5 results



EVALUATION

Measured results show strong label alignment, weak separation

A. 2,127 Articles **B. 0.032** Silhouette (cosine) **C. 0.92** Adjusted Rand Index **D. 0.89** Normalized Mutual Information

CLUSTER SIZES

Cluster	Articles
Labour / Election / Blair	407
Game / Win / Cup	505
Growth / Shares / Company	495
Film / Awards / Best	373
Users / Technology / Software	347

SILHOUETTE BY K

k	Silhouette
5	0.0324
6	0.0323
8	0.0342
10	0.0361

Takeaway: Silhouette is low and nearly unchanged across k, so keywords matter too. ARI and NMI are after-training checks, not accuracy. 5 automated tests pass.

BOUNDARIES AND NEXT STEPS

Explainability is the strength And the boundary

LIMITATIONS

- TF-IDF ignores word meaning.
- K-Means needs k in advance.
- 2D PCA map keeps about 2.4% of the information.
- Dataset is from 2004–2005 with only 5 broad areas.

FUTURE WORK

- Try LDA, NMF, or sentence embeddings.
- Choose k automatically.
- Add upload-your-own-CSV support.
- Use a larger and newer dataset.

Conclusion: A simple, explainable, reproducible system that discovers news topics without labels and runs on an ordinary laptop.